

Jevons' Paradox as a Safety-Critical Control Problem: Consistent Rebound Dynamics and the Feasibility Limits of Fiscal Feedback

Peter Zsoldos

Department of Computer Science and Electrical Engineering
Tampere University, Tampere, Finland
zsoldosp44@gmail.com

Abstract—Efficiency gains can raise total resource use—Jevons' paradox—which invites a control formulation in which a dynamic resource tax rejects efficiency shocks as disturbances. We show that the natural two-state encoding of this idea, carrying resource draw N and output Y as independent states alongside the definition $e = Y/N$ and an exogenous efficiency path, is over-determined, is not invariant to the units of the efficiency index, and structurally cannot represent efficiency savings, so backfire is assumed rather than derived. Imposing the identity yields a scalar plant that is linear in the geodesic coordinate $x = \ln N$ and affine in the log gross tax $v = \ln(1 + \tau)$ with constant actuation gain. In this plant backfire holds *iff* the demand elasticity satisfies $\eta > 1$, and efficiency lowers consumption on impact before raising it. Two conclusions reverse: the disturbance that matters is the efficiency level, not its velocity, which is stabilising; and first-order Taylor prediction of delayed data yields a neutral-type loop with exact margin $\kappa\tau_d < 0.9417$, below the $\pi/2$ margin of the unpredicted loop. The main result is a feasibility obstruction: autonomous demand growth alone needs a 75% peak tax and is controllable, whereas adding the rebound channel raises the requirement to 223% and makes barrier feasibility fail for $\eta > 0.948$ —stricter than the backfire threshold. Feedback cuts cumulative overshoot by 68% but cannot render the safe set invariant at credible tax ceilings.

Index Terms—Control barrier functions, predictor feedback, neutral delay systems, actuator saturation, resource economics, rebound effect.

I. INTRODUCTION

William Stanley Jevons observed in 1865 that efficiency gains in coal use increased, rather than reduced, aggregate coal consumption [1]. When the economy-wide rebound exceeds 100% the condition is called backfire [2]–[4]. Its magnitude is contested: Gillingham *et al.* [7] find little support for economy-wide backfire, while sectoral ICT studies report rebound factors of 1.2–1.6 [6], [8]. A useful model must therefore treat the rebound magnitude as a parameter and report how conclusions depend on it—the approach taken here.

A natural idea is to treat resource policy as feedback control: make the tax a function of the measured state, so efficiency shocks are rejected as disturbances rather than absorbed as consumption growth. We pursued that programme and report that it survives only after substantial repair. The two-state encoding of Section II-A is our own first formulation, implemented and calibrated before we found the defects reported below; we present it rather than quietly discarding

it because the defects are ones any first attempt is likely to reproduce, and the corrected model reverses two conclusions we had drawn from the flawed one.

Related work. Rebound quantification dates to the Khazzoom–Brookes postulate [2], [3], surveyed in [5], [7]; computable general equilibrium approaches are descriptive, not suited to algorithmic policy design. Ecological economists argue for absolute caps [9], and the choice between a price and a quantity instrument is Weitzman's classical problem [10]: quantity dominates when marginal damage is steep relative to marginal abatement—precisely the case at a hard threshold, of which our feasibility result is a dynamic restatement. On the control side, forward invariance of a safe set is the province of control barrier functions [12], not control Lyapunov functions [11]—often conflated, and separated in Section III. For delay compensation the lineage is the dead-time compensator [13], finite spectrum assignment [14] and predictor feedback [15]; stability of the resulting neutral equations needs the difference-operator machinery of [16] and the eigenvalue methods of [17].

Contributions. (i) Three independent defects in the natural two-state formulation (Section II-A). (ii) A consistent scalar plant (Section II-B) in which backfire is derived, not assumed, and which reproduces the observed Jevons time signature. (iii) A corrected delay-margin theorem (Section IV): the first-order Taylor predictor's exact margin is *smaller* than the unpredicted loop's. (iv) A feasibility obstruction (Sections III–V): because the required tax is convex in the log-consumption gap, the rebound channel is the increment that crosses the actuator bound despite not being the larger contributor. (v) Negative findings reported rather than suppressed (Section V-A).

II. MODEL

A. Three Defects of the Two-State Formulation

Our first formulation was the one that suggests itself immediately: state $\mathbf{x} = [N, Y]^T$, efficiency $e = Y/N$ declared as a definition, an exogenously prescribed efficiency path $e(t)$, and dynamics

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\tau + \mathbf{D}\dot{e}(t), \quad (1)$$

with logistic f , multiplicative g and constant coupling $\mathbf{D}$. We implemented and calibrated (1) against IEA data-centre electricity projections [19]. It has three independent defects.

(D1) *Over-determination*. If $e = Y/N$ is an identity and N, Y each carry an independent ODE, $e(t)$ is determined by the state and cannot also be prescribed; the prescribed $e(t)$ and the realised $Y(t)/N(t)$ diverge by up to $6.4\times$ over 40 yr. (D2) *Units dependence*. The rebound enters as $D_N \dot{e}$, not invariant under $e \mapsto ce$: rescaling the efficiency index—physically the same technology in different units—moves peak consumption from 3.96 to 7.43 normalised units. A prediction depending on the units of an index is not a prediction. (D3) *No savings channel*. With $D_N > 0$ and $\dot{e} > 0$ every efficiency gain strictly increases N ; no parameter gives $\partial N / \partial e < 0$, so backfire is assumed in (1) rather than found.

These defects are structural, not calibration issues; every quantitative claim derived from (1) must be regarded as an artefact. The plant below is a replacement, not a refinement, and all numbers in Section V are recomputed from scratch.

B. A Consistent Rebound Plant

We keep the object that is genuinely exogenous—technology—and impose the identity rather than duplicating it. Let $\mu(t) := \ln e(t)$ be log efficiency. Because $e \mapsto ce$ shifts μ by a constant, any model depending on μ only through differences and derivatives is scale-free, curing D2. Let Y denote demand for the energy *service*, with effective service price $p_{\text{eff}}(t) = (1 + \tau(t))p_0/e(t)$, so that efficiency lowers and the tax raises the cost of a unit of service. With constant-elasticity long-run demand and partial adjustment at rate $\lambda > 0$,

$$\begin{aligned} \frac{d}{dt} \ln Y &= \lambda (\ln Y^*(t) - \ln Y(t)), \\ \ln Y^*(t) &= \ln A_0 + \gamma t + \eta \mu(t) - \eta v(t), \end{aligned} \quad (2)$$

where $\eta > 0$ is the demand (rebound) elasticity, γ the autonomous income growth rate of service demand, and $v := \ln(1 + \tau) \in \mathcal{V} = [0, v_{\max}]$, $v_{\max} = \ln(1 + \tau_{\max})$, is the *log gross tax*. Imposing $N = Y/e$, i.e. $x := \ln N = \ln Y - \mu$, gives the scalar plant

$$\dot{x} = \lambda \underbrace{[\ln A_0 + \gamma t + (\eta - 1)\mu(t) - x]}_{=: F(x,t)} - \dot{\mu}(t) + Gv, \quad (3)$$

with *constant* actuation gain $G = -\lambda \eta < 0$. We normalise $\mu(0) = 0$, which is free by the scale-invariance above, and set $\ln A_0 = \ln N(0)$ so that the untaxed system starts on its frozen-time equilibrium; all boundary pressure in what follows is therefore generated by γ and μ rather than by an initial transient.

Three structural gains follow. The plant is exactly affine in the control; the actuation gain is constant, so the singularity $g_N(\mathbf{x}) = 0$ that afflicts (1) cannot occur and no policy-irrelevance caveat is needed; and (3) is *linear* in x , so the delay analysis of Section IV is exact rather than a linearisation. The accessibility distribution of (1) degenerates on a codimension-one locus; once the identity is imposed the system is one-dimensional with $G \neq 0$ everywhere, so that apparent geometric

richness was an artefact of the spurious degree of freedom introduced by D1.

C. Backfire is Derived, Not Assumed

Proposition 1 (Backfire condition). *Let μ and v be constant. The frozen-time equilibrium of (3) satisfies $x_{\text{ss}} = \ln A_0 + \gamma t + (\eta - 1)\mu - \eta v$, so $\partial x_{\text{ss}} / \partial \mu = \eta - 1$. A permanent efficiency improvement therefore raises long-run consumption iff $\eta > 1$, leaves it unchanged iff $\eta = 1$, and lowers it iff $\eta < 1$.*

Proof. Set $\dot{x} = 0$ and $\dot{\mu} = 0$ in (3), solve for x and differentiate in μ . Along the ramp γt the solution tracks x_{ss} with the constant offset $-\gamma/\lambda$, which is independent of μ and therefore does not affect $\partial x_{\text{ss}} / \partial \mu$. $\square$

Proposition 2 (Jevons time signature). *The instantaneous response of consumption growth to an efficiency impulse is $\partial \dot{x} / \partial \dot{\mu} = -1 < 0$ for every η , while the long-run response is $\eta - 1$. Efficiency therefore always reduces consumption on impact, and the rebound arrives with the demand adjustment time constant $1/\lambda$.*

Proof. Read the coefficient of $\dot{\mu}$ in (3) and combine with Proposition 1. $\square$

Proposition 1 recovers the textbook rebound threshold as a property of the dynamics rather than an input, curing D3; the quasi-static target x_{ss} carries the efficiency contribution $(\eta - 1)\mu(t)$ exactly, by construction, and we verified this identity numerically across $\eta \in [0.4, 1.8]$, where $M := \mu(T) - \mu(0)$. Proposition 2 is the content of Fig. 1(a), which plots the *realised* trajectory rather than this quasi-static target: for $\eta = 1.4$ the efficiency contribution dips to -0.42 near $t = 13$ yr, then recovers to only $+0.68$ by the horizon $t = 25$ yr, short of the quasi-static $(\eta - 1)M = 0.88$ by the demand-adjustment lag $1/\lambda = 4$ yr—so the 25-year window in Table I understates the eventual backfire rather than overstating it. The two-state model (1) cannot produce this sign reversal at any parameter value. Fig. 1(b) sweeps η and locates the two thresholds that organise the paper: the backfire condition $\eta = 1$ and the feasibility limit $\eta_{\text{crit}} = 0.948$ of Corollary 1.

Remark 1 (The disturbance is the level, not the velocity). In (3) the efficiency level enters with coefficient $\lambda(\eta - 1)$ and its velocity with coefficient -1 . The velocity term is therefore *stabilising*: rapid innovation transiently relieves rather than aggravates boundary pressure. This reverses the premise that $\dot{e}$ is the disturbance to be rejected. The quantity that must be tracked is the accumulated level $(\eta - 1)\mu(t)$.

III. GEOMETRY, CONTROL DESIGN AND FEASIBILITY

Let the safe set be $\Omega = \{x : h(x) \geq 0\}$ with $h(x) := x^* - x$ and $x^* = \ln N^*$, and take the quadratic Lyapunov function $V(x) = \frac{1}{2}(x - x^*)^2$ in the log coordinate: $x = \ln N$ is where the plant (3) is linear, and it is also the geodesic coordinate of the metric $g_{NN} = 1/(\sigma N^2)$ for an elasticity of substitution σ , which enters V only as a positive constant and so cancels from $\dot{V} \leq -KV$ —useful, since σ is among the least reliably identified parameters

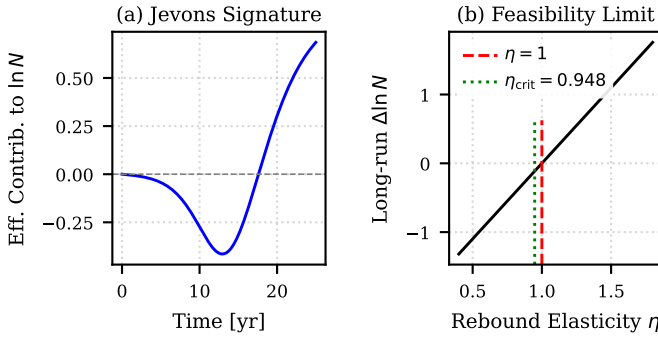


Fig. 1. Corrected rebound mechanism. (a) The efficiency contribution to $\ln N$ is negative on impact and positive in the long run, reversing sign near $t = 17.6$ yr for $\eta = 1.4$; the two-state model cannot produce this. (b) Long-run consumption versus rebound elasticity, with the derived backfire threshold $\eta = 1$ and the feasibility limit $\eta_{\text{crit}} = 0.948$.

TABLE I
CALIBRATION (N NORMALISED BY $N^* = 460$ TWH/YR)

Symbol	Meaning	Value	Basis
N^*	Resource ceiling	1.0	IEA high scenario [19]
$N(0)$	2023 data-centre draw	0.804	IEA [19]
A_0	Demand scale	0.804	$x(0)$ on frozen-time eq.
$\mu(0)$	Efficiency index origin	0	normalisation (D2)
λ	Demand adjustment rate	0.25	≈ 2.8 yr half-life
γ	Autonomous service growth	0.04	long-run digital demand
η	Rebound elasticity	1.40	ICT range [6], [8]
M	Total log efficiency gain	$\ln 9$	three GPU generations
τ_{max}	Actuator ceiling	1.0	100% tax
τ_d	Reporting delay	1.0 yr	macro release lag
κ	Lyapunov gain	2.0	design choice
T	Horizon	25 yr	2025–2050

in applied production theory. Also $V \rightarrow \infty$ as $N \rightarrow 0^+$, so positivity of the resource state is preserved structurally.

A. Lyapunov Tracking and Barrier Filtering

Imposing $\dot{V} \leq -KV$ on (3) and writing $z := x - x^*$ gives, in the interior of $\mathcal{V}$,

$$v_{\text{CLF}} = \left[\frac{F(x, t) + \kappa z}{-G} \right]_0^{v_{\text{max}}}, \quad \kappa := \frac{K}{2}, \quad (4)$$

where $[\cdot]_0^{v_{\text{max}}}$ denotes clipping to $\mathcal{V}$; the unsaturated closed loop is $\dot{z} = -\kappa z$. The factor $\kappa = K/2$ follows from $\dot{V} \leq -KV$ with $V = \frac{1}{2}z^2$; all delay conditions below are stated in κ , and expressing them in K introduces a factor-of-two error.

A Lyapunov function certifies convergence to a target; it does not certify that a boundary is never crossed. With class- $\mathcal{H}$ gain $\alpha > 0$, the invariance condition $\dot{h} \geq -\alpha h$ gives, since $G < 0$,

$$v \geq v_{\text{CBF}}(x, t) := \frac{F(x, t) - \alpha(x^* - x)}{-G}, \quad (5)$$

and we deploy the minimum-intervention filter $v^* = [v_{\text{CBF}}(x, t)]_0^{v_{\text{max}}}$. Far below the ceiling h is large and (5) is slack, so $v_{\text{CBF}} < 0$ and no tax is levied: the nominal policy is no tax at all and the barrier is the only thing that ever raises τ . If $x(0) \in \Omega$ and $v_{\text{CBF}}(x(t), t) \leq v_{\text{max}}$ for all $t \geq 0$, then v^*

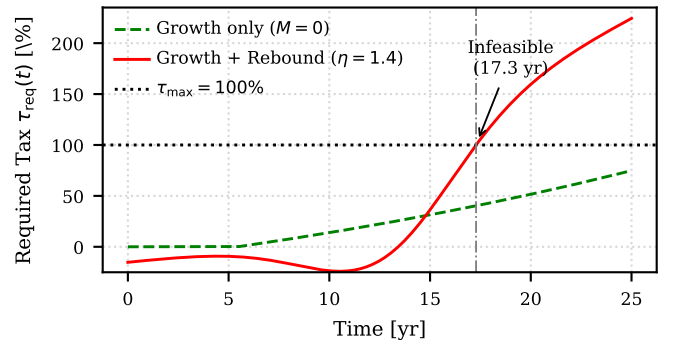


Fig. 2. The central result. Income growth alone requires a peak tax of 75% and never exceeds the credible ceiling. Adding the rebound channel raises the requirement to 223% and makes the invariance certificate infeasible from $t = 17.3$ yr. The early dip below zero is the transient savings channel of Proposition 2.

satisfies $\dot{h} \geq -\alpha h$ pointwise, so $h(t) \geq h(0)e^{-\alpha t} \geq 0$ by the comparison lemma and Ω is forward invariant [12].

B. The Feasibility Obstruction

Evaluating (5) on $\partial\Omega$ defines the tax required to hold the ceiling,

$$v_{\text{req}}(t) = \frac{F(x^*, t)}{-G} = \frac{\lambda [\ln A_0 + \gamma t + (\eta - 1)\mu(t) - x^*] - \dot{\mu}(t)}{\lambda \eta}, \quad (6)$$

with $\tau_{\text{req}} = e^{v_{\text{req}}} - 1$.

Proposition 3 (Admissibility). Ω admits an invariance certificate at time t iff $v_{\text{req}}(t) \leq v_{\text{max}}$. Under Table I this fails on 30.8% of the horizon, first at $t = 17.3$ yr, with $\max_t \tau_{\text{req}} = 223\%$. Setting $M = 0$ (income growth only) gives $\max_t \tau_{\text{req}} = 75\%$ and the condition never fails.

The decomposition behind Proposition 3 is the paper's central quantitative point. At the horizon the log-consumption gap the tax must close is $\underbrace{\gamma T}_{1.000(53\%)} + \underbrace{(\eta - 1)M}_{0.879(47\%)} + \ln A_0$, so the two drivers are of comparable size. But v_{req} is linear in the gap while $\tau_{\text{req}} = e^{v_{\text{req}}} - 1$ is exponential in it: a gap of 0.78 needs 75%, a gap of 1.66 needs 223%. The rebound channel is not the larger contributor, yet it is the increment that carries the requirement across the actuator bound, because the marginal cost of closing the gap rises exponentially (Fig. 2).

Remark 2 (Convexity of the actuation requirement). The mechanism above is convexity, not arithmetic, with a consequence for how the requirement should be reported. Write $\Gamma(t) := \ln A_0 + \gamma t + (\eta - 1)\mu(t) - x^*$ for the log-consumption gap, so v_{req} is affine in Γ while $\tau_{\text{req}} = e^{v_{\text{req}}} - 1$ is strictly convex in it. So attributing infeasibility to a single channel is meaningful only at the margin, as in Proposition 3; and if Γ is uncertain, Jensen's inequality gives $\mathbb{E}[\tau_{\text{req}}] \geq e^{\mathbb{E}[v_{\text{req}}]} - 1$, with equality only for a degenerate distribution—the tax required on average strictly exceeds the tax at the nominal calibration, and the gap widens with the dispersion of (η, γ, M) . An actuator

ceiling sized at nominal values is therefore systematically under-sized; this argues for reporting the distribution of τ_{req} rather than a point value.

Corollary 1 (Critical elasticity). *At $\tau_{\text{max}} = 1$ the condition of Proposition 3 holds for all t iff $\eta \leq \eta_{\text{crit}} = 0.948$. Since $\eta_{\text{crit}} < 1$, controllability is lost before backfire sets in: the policy-relevant threshold is stricter than the economic one.*

Proof. Since η enters (6) both as a multiplier of $\mu(t)$ and as the divisor, monotonicity in η is not immediate and deserves an argument rather than an appeal to Fig. 1(b) alone. Write

$$v_{\text{req}}(\eta, t) = \mu(t) + \frac{\Psi(t)}{\lambda\eta}, \quad (7)$$

$$\Psi(t) := \lambda [\ln A_0 + \gamma - x^* - \mu(t)] - \dot{\mu}(t),$$

obtained by expanding (6); Ψ does not depend on η . Hence for fixed t , $v_{\text{req}}(\cdot, t)$ is monotonic on $(0, \infty)$ with $\partial v_{\text{req}}/\partial \eta = -\Psi(t)/(\lambda\eta^2)$, strict whenever $\Psi(t) \neq 0$. As the efficiency path saturates, $\dot{\mu}(t), \ddot{\mu}(t) \rightarrow 0$ near $t = T$, so $\partial v_{\text{req}}/\partial t = \dot{\mu}(t) + \dot{\Psi}(t)/(\lambda\eta) \rightarrow \gamma/\eta > 0$ there for every $\eta > 0$: the requirement is eventually increasing in t regardless of elasticity. Under Table I this terminal regime governs the maximum, $g(\eta) := \max_{t \in [0, T]} v_{\text{req}}(\eta, t) = v_{\text{req}}(\eta, T) = \mu(T) + \Psi(T)/(\lambda\eta)$, throughout $\eta \in [0.7, 2.0] \ni \eta_{\text{crit}}$, which we confirm numerically; for smaller η the interior maximum occurs during the transient dip of Proposition 2 and is negative, hence trivially below v_{max} . Since $\Psi(T) < 0$ at this calibration, g is strictly increasing on $[0.7, 2.0]$, so it crosses v_{max} exactly once; η_{crit} is that crossing, and $g(\eta) \leq v_{\text{max}}$ holds iff $\eta \leq \eta_{\text{crit}}$. $\square$

Lemma 1 (ISS with respect to estimate error). *Let the terms of F be estimated with total error bounded by δ and let the actuator be unsaturated. Then $|z(t)| \leq |z(0)|e^{-\kappa t} + \delta/\kappa$.*

Proof. The perturbed loop is $\dot{z} = -\kappa z + w$ with $|w| \leq \delta$; apply the comparison lemma to $\frac{d}{dt}|z| \leq -\kappa|z| + \delta$. $\square$

Lemma 1 holds only while $v \in (0, v_{\text{max}})$. Once $v \equiv v_{\text{max}}$ the loop is open and no input-to-state bound applies—a caveat binding for roughly 30% of our horizon.

IV. DELAY COMPENSATION

Macroeconomic aggregates are released with a lag $\tau_d \approx 0.5$ –1.5 yr. The natural first attempt extrapolates one delay interval forward, $\hat{x}(t) = x(t - \tau_d) + \tau_d \dot{x}(t - \tau_d)$, giving the closed-loop error equation

$$\dot{z}(t) = -\kappa[z(t - \tau_d) + \tau_d \dot{z}(t - \tau_d)], \quad (8)$$

which contains the delayed *derivative* and is therefore of neutral type [16]. Equivalently $\frac{d}{dt}[z(t) + \kappa\tau_d z(t - \tau_d)] = -\kappa z(t - \tau_d)$, whose difference operator is stable iff $\kappa\tau_d < 1$. Since (3) is exactly linear, (8) is the exact closed loop.

Theorem 1 (Exact delay margin of the Taylor predictor). *Let $\kappa > 0$, $\tau_d > 0$ and $h := \kappa\tau_d$. The loop (8) is exponentially stable iff*

$$h < h^* = \frac{v^*}{\sin v^* (1 + (v^*)^2)} \approx 0.941680, \quad (9)$$

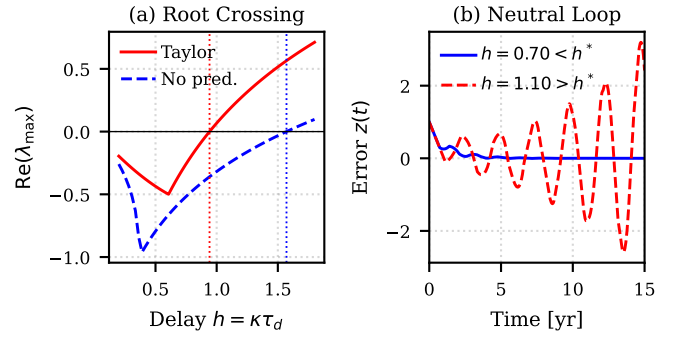


Fig. 3. Corrected delay analysis. (a) Rightmost characteristic root versus $h = \kappa\tau_d$: the first-order Taylor predictor loses stability at $h^* \approx 0.942$, before the unpredicted loop at $\pi/2$. (b) Simulation of the exactly linear plant without saturation.

where $v^* \approx 2.798386$ is the smallest positive root of $\cos v + v \sin v = 0$.

Proof. The characteristic function is $\Delta(s) = s + \kappa(1 + \tau_d s)e^{-s\tau_d}$. With $\lambda_c = s\tau_d$ and $h = \kappa\tau_d$ this is $\lambda_c + h(1 + \lambda_c)e^{-\lambda_c} = 0$. As $h \rightarrow 0^+$ the finite root is $\lambda_c \approx -h < 0$, so the loop begins stable and instability requires an imaginary-axis crossing. Put $\lambda_c = jv$, $v > 0$. Using $(1 + jv)(\cos v - j \sin v) = (\cos v + v \sin v) + j(v \cos v - \sin v)$,

$$\text{Re} : h(\cos v + v \sin v) = 0, \quad (10)$$

$$\text{Im} : v + h(v \cos v - \sin v) = 0. \quad (11)$$

As $h > 0$, (10) forces $\cos v = -v \sin v$; substituting into (11) gives $h = v/(\sin v - v \cos v) = v/[\sin v(1 + v^2)]$. The smallest positive root of $\cos v + v \sin v = 0$ is v^* , giving h^* ; larger roots give larger h and so do not govern the first crossing. Since $h^* < 1$ the difference operator is stable throughout $[0, h^*)$, no root chain escapes to $+\infty$, and the root condition is necessary and sufficient [16], [17]. $\square$

Proposition 4 (No prediction beats first-order prediction). *Feeding back the delayed measurement without prediction gives $\dot{z}(t) = -\kappa z(t - \tau_d)$, exponentially stable iff $\kappa\tau_d < \pi/2$. The Taylor predictor therefore reduces the delay margin by the factor $\pi/(2h^*) \approx 1.668$, a loss of about 40%.*

Proof. For $\Delta(s) = s + \kappa e^{-s\tau_d}$ and $s = j\omega$ the real part gives $\cos \omega\tau_d = 0$, so $\omega\tau_d = \pi/2$, and the imaginary part gives $\omega = \kappa$. Compare with Theorem 1. $\square$

Both constants were confirmed by numerical continuation of the rightmost characteristic root; the computed sign changes bracket 0.9417 and 1.5708 (Fig. 3(a)); Fig. 3(b) shows the corresponding unsaturated trajectories on either side of h^* .

Remark 3 (Why a retarded Lyapunov–Krasovskii argument fails). It is tempting—and we initially attempted—to certify (8) with $V_{LK} = \frac{1}{2}z^2(t) + \frac{\kappa\tau_d}{2} \int_{t-\tau_d}^t z^2(\theta) d\theta$ and conclude $\tau_d < 1/\kappa$. This fails, and worth recording because the bound errs in *direction*, not merely *tightness*: a memory term penalising z^2 cannot dominate the cross term in $\dot{z}(t - \tau_d)$ that a *neutral*

TABLE II
CONTROLLER PERFORMANCE ($\eta = 1.4$, $\tau_d = 1$ yr, $\tau_{\max} = 1$, $T = 25$ yr)

Controller	peak N	IAE	ISCE	Y_f	S [%]	breach
Open loop	3.682	14.30	0.00	32.93	0.0	yes
Static tax 22%	2.789	12.09	1.21	24.94	0.0	yes
CLF, no lag	1.543	4.64	8.55	13.80	30.3	yes
CLF + Taylor pred.	1.542	4.65	8.55	13.79	29.6	yes
CLF + Smith pred.	1.542	4.63	8.56	13.79	30.3	yes
CBF filter + Smith pred.	1.538	4.65	8.59	13.76	30.0	yes

equation produces—a functional penalising $\dot{z}^2$ over the window is required [16]—and closing the argument by asserting $z^2(t - \tau_d) \geq \tau_d^2 \dot{z}^2(t - \tau_d)$ is circular, since along $\dot{z} = -\kappa z$ it reads $1 - (\kappa \tau_d)^2 \geq 0$, the conclusion itself. The resulting $\kappa \tau_d < 1$ coincides with the (necessary but weaker) difference-operator condition, not h^* . A linear delay equation does not undergo a Hopf bifurcation; it exhibits a root crossing.

The remedy is the classical dead-time compensator [13]–[15]: integrate the model across the unmeasured window rather than extrapolating linearly,

$$\hat{x}(t) = x(t - \tau_d) + \int_{t-\tau_d}^t [F(\hat{x}, s) + Gv(s)] ds, \quad (12)$$

evaluated by fixed-step RK4 using the known applied input history. Under an exact model the prediction is exact, the delay is cancelled and $\dot{z} = -\kappa z$ is recovered for arbitrary τ_d . An innovation term $L(x(s) - \hat{x}(s))$ may be appended to (12) only for $s \leq t - \tau_d$; inside the prediction window no measurement exists, so the integration there must be open loop. Implementations that interpolate the true state across the window are non-causal. Enforcing strict causality changes reported metrics by less than 0.2% under a well-matched model, but the distinction is required for correctness and would matter under larger mismatch.

V. NUMERICAL RESULTS

Equation (3) was integrated with adaptive RK45 ($\text{rtol} = 10^{-7}$, $\text{atol} = 10^{-9}$) over $T = 25$ yr under Table I. Metrics are peak N ; integrated absolute error $\text{IAE} = \int |N - N^*| dt$; integrated squared effort $\text{ISCE} = \int \tau^2 dt$; terminal output Y_f ; saturation duty S ; and whether the trajectory breaches $1.05N^*$.

Attenuation without invariance. Table II gives the primary comparison: closed-loop feedback reduces IAE from 14.30 to 4.63 (67.6%) and peak consumption from $3.682N^*$ to $1.542N^*$ (58.1%)—both large and policy-relevant, but neither is invariance: every admissible controller breaches the ceiling, consistent with Proposition 3, with the actuator saturated for about 30% of the horizon (Fig. 4).

Predictor comparison. At the nominal calibration $\kappa \tau_d = 1.0$, only marginally beyond h^* , the two predictors are close in Table II; Table III separates them in the unsaturated regime: the Taylor predictor’s residual oscillation grows by a factor of 768 across the sweep, inflecting sharply at $\kappa \tau_d \approx h^*$ as Theorem 1 predicts, against 4.6 for the model-based predictor—inverting the intuition, which we initially followed, that first-order prediction enlarges the stability region.

TABLE III
RESIDUAL OSCILLATION OF N VS. REPORTING DELAY ($\kappa = 1$, UNSATURATED)

τ_d [yr]	$\kappa \tau_d$	Taylor pred.	Smith pred.
0.50	0.50	0.0011	0.0008
0.90	0.90	0.0034	0.0013
1.00	1.00	0.0066	0.0014
1.25	1.25	0.0439	0.0019
1.50	1.50	0.3451	0.0024
2.00	2.00	0.8310	0.0038

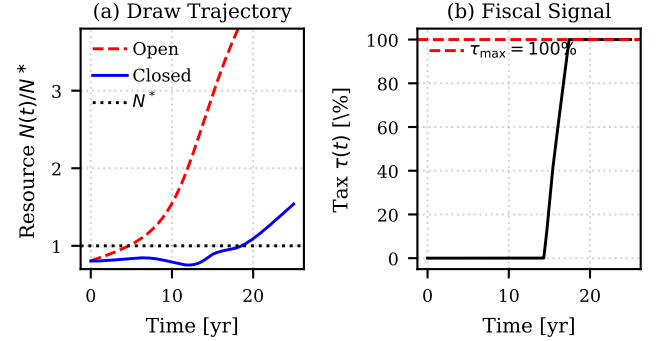


Fig. 4. Trajectories at $\tau_d = 1$ yr. (a) Resource draw: feedback delays and flattens the breach but does not prevent it. (b) Fiscal signal: the tax is railed at $\tau_{\max}$ over the final third of the horizon, which is why controller-design choices have little leverage.

Required authority and its cost. Sweeping the actuator ceiling, invariance first obtains at $\tau_{\max} \approx 220\%$ —close to the analytic 223% of Proposition 3—at a cost of 35.0% of terminal output ($13.76 \rightarrow 8.94$): defending the 460 TWh/yr ceiling against these paths costs roughly a third of terminal digital output.

Robustness. Over 300 Monte Carlo draws of the four genuinely uncertain quantities $(\eta, \lambda, \gamma, M)$, the open loop breaches in 94% of draws and the closed loop in 77%; no draw destabilises, but the residual breach rate is governed by whether $\eta > 1$, not control design. Consistent with Remark 2, the same draws give a mean peak requirement $\mathbb{E}[\max_t \tau_{\text{req}}] = 241\%$ against the nominal 223% and a 90th percentile of 318%: sizing $\tau_{\max}$ from the nominal calibration understates the requirement at every quantile above the median.

A. Negative Findings

Several comparisons one might expect to favour the proposed machinery do not: with a non-negative tax the barrier and Lyapunov designs coincide almost exactly (peak 1.538 vs. 1.542; Y_f 13.76 vs. 13.79) and, with ample authority, both converge onto the ceiling ($N = 0.9994$ and 0.9996)—we had expected the barrier to preserve a margin, but the disturbance drives N upward and a one-sided tax can only resist. Varying the class- $\mathcal{K}$ gain α over $[0.2, 2.0]$ moves peak N only from 1.519 to 1.544, and holding the controller’s internal model at nominal while perturbing the plant changes the 95th-percentile peak from 1.887 to 1.890. All three share one explanation: for roughly a third of the horizon the actuator is railed, and

a railed actuator consults neither its model, its gain, nor its certificate—itself a reason to report the feasibility boundary rather than a controller ranking. The barrier’s value here is the *certificate* (6), not improved trajectories.

VI. DISCUSSION AND CONCLUSION

Treating resource policy as feedback is well posed and worthwhile: it delivers a 68% reduction in cumulative boundary violation and a 58% reduction in peak draw from a signal computable from delayed data, provided the predictor is model-based rather than a truncated expansion. What it does not buy is the elimination of backfire. The obstruction is structural: because the required tax is exponential in the log-consumption gap, a price instrument capable of defending a hard ceiling against sustained demand growth plus rebound must be so large that its political and distributional feasibility becomes the binding constraint instead—the dynamic counterpart of Weitzman’s result [10] that, near a steep threshold, quantity instruments dominate price instruments. The constructive reading is that $v_{\text{req}} - v_{\text{max}}$ in (6) is computable in real time from observable data and is precisely the signal indicating when a price instrument must be supplemented by a quantity instrument.

Two reversals relative to the velocity-rejection view deserve emphasis: by Remark 1 the disturbance to be tracked is the efficiency *level*, not its velocity, so a controller built to reject $\dot{e}$ is rejecting a stabilising signal; and by Proposition 4 the intuitive delay fix makes matters worse. Both errors are invisible in a model with defects D1–D3—the practical argument for dimensional and definitional consistency before control design.

Limitations. Equation (3) is a scalar reduced form with one instrument—no capital stock, sectoral disaggregation, trade leakage, endogenous innovation response, or forward-looking expectations. The last is most serious: an agent anticipating a rising tax would reallocate intertemporally, a channel a partial-adjustment rule cannot represent. The headline depends on $\eta > 1$, which the economy-wide literature does not establish [7]; Corollary 1 and the full sweep, not a point claim, are the appropriate summary. We model a pure reporting delay rather than quarterly sampling, which would add roughly half a sampling interval of effective delay and require discrete-time barrier conditions; Y_f is a scalar output proxy with no welfare or incidence analysis. Future work should replace the pointwise barrier with a receding-horizon formulation [18] over a forecast μ trajectory, treat τ_{max} as a decision variable in a joint price–quantity design, and identify η and λ from data with quantified uncertainty, since the policy conclusion turns on which side of 0.948 the elasticity lies.

Jevons’ paradox is a legitimate object of control-theoretic study, but the obvious state-space encoding of it is not. Imposing the identity yields a scalar plant in which backfire holds iff $\eta > 1$ and efficiency gains lower consumption before raising them. In that plant delay compensation by first-order Taylor extrapolation *shrinks* the delay margin from $\kappa\tau_d < \pi/2$ to $\kappa\tau_d < 0.9417$. Autonomous demand growth alone is controllable with a 75% tax; adding the rebound

channel raises the requirement to 223%, and the feasibility limit $\eta_{\text{crit}} = 0.948$ is stricter than the backfire threshold. Fiscal feedback is therefore necessary for boundary defence, demonstrably insufficient on its own, and the safety-critical formulation makes that insufficiency measurable in advance.

ACKNOWLEDGMENT

Portions of this manuscript were drafted with the assistance of a large language model (Anthropic Claude), which was used for prose drafting and L^AT_EX preparation in Sections II–A–V–A; the model formulation, proofs, numerical results and all conclusions were derived and verified by the author, who takes full responsibility for the content.

REPRODUCIBILITY

Code and calibration for every figure and table are at <https://github.com/Hannibal0319/jevons-paradox-control>: `tables.py` reproduces Tables II–III to the digit, `figures.py` reproduces Figs. 1–4. Propositions 1–2 and Corollary 1 were checked against simulation; the delay margins of Theorem 1 and Proposition 4 against numerical continuation of the rightmost characteristic root.

REFERENCES

- [1] W. S. Jevons, *The Coal Question*. London, UK: Macmillan, 1865.
- [2] J. D. Khazzoom, “Economic implications of mandated efficiency in standards for household appliances,” *Energy J.*, vol. 1, no. 4, pp. 21–40, 1980.
- [3] L. Brookes, “Energy efficiency and economic growth,” *Energy Policy*, vol. 18, no. 2, pp. 199–201, 1990.
- [4] H. D. Saunders, “Fuel conserving (and using) production functions,” *Energy Econ.*, vol. 30, no. 5, pp. 2184–2235, 2008.
- [5] S. Sorrell, “Jevons’ paradox revisited: The evidence for backfire from improved energy efficiency,” *Energy Policy*, vol. 37, no. 4, pp. 1456–1469, 2009.
- [6] S. Sorrell, *The Rebound Effect*. London, UK: UK Energy Research Centre, 2007.
- [7] K. Gillingham, D. Rapson, and G. Wagner, “The rebound effect and energy efficiency policy,” *Rev. Environ. Econ. Policy*, vol. 10, no. 1, pp. 68–88, 2016.
- [8] V. C. Coroamă and F. Mattern, “Digital rebound: Why digitalization will not redeem us our environmental sins,” in *Proc. ICT4S*, Lappeenranta, Finland, 2019.
- [9] J. Rockström *et al.*, “A safe operating space for humanity,” *Nature*, vol. 461, no. 7263, pp. 472–475, 2009.
- [10] M. L. Weitzman, “Prices vs. quantities,” *Rev. Econ. Stud.*, vol. 41, no. 4, pp. 477–491, 1974.
- [11] P. Kokotović and M. Arcak, “Constructive nonlinear control: A historical perspective,” *Automatica*, vol. 37, no. 5, pp. 637–662, 2001.
- [12] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, “Control barrier function based quadratic programs for safety critical systems,” *IEEE Trans. Autom. Control*, vol. 62, no. 8, pp. 3861–3876, 2017.
- [13] O. J. M. Smith, “Closer control of loops with dead time,” *Chem. Eng. Prog.*, vol. 53, no. 5, pp. 217–219, 1957.
- [14] A. Z. Manitius and A. W. Olbrot, “Finite spectrum assignment problem for systems with delays,” *IEEE Trans. Autom. Control*, vol. 24, no. 4, pp. 541–552, 1979.
- [15] M. Krstić, *Delay Compensation for Nonlinear, Adaptive, and PDE Systems*. Boston, MA: Birkhäuser, 2009.
- [16] J. K. Hale and S. M. Verduyn Lunel, *Introduction to Functional Differential Equations*. New York, NY: Springer-Verlag, 1993.
- [17] W. Michiels and S.-I. Niculescu, *Stability and Stabilization of Time-Delay Systems: An Eigenvalue-Based Approach*. Philadelphia, PA: SIAM, 2007.
- [18] D. Q. Mayne, “Model predictive control: Recent developments and future promise,” *Automatica*, vol. 50, no. 12, pp. 2967–2986, 2014.
- [19] International Energy Agency, *Electricity 2024: Analysis and Forecast to 2026*. Paris, France: IEA, 2024.